Parth Sachin Deshmukh

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Education

Texas A&M University, College Station, TX Master of Science in Computer Science May 2027
Coursework: Artificial Intelligence, Analysis of Algorithms, Data Mining

Vishwakarma Institute of Information Technology, Pune, India Bachelor of Technology in Information Technology — GPA: 9.12 / 10.00 May 2024

Skills

Languages: C++, Python, SQL, JavaScript, R
Build & DevOps: CMake, scikit-build-core, Docker, Git/GitHub, CI/CD, Conda, PyPI Packaging
AI & ML: LangChain, LLMs, Retrieval-Augmented Generation, TensorFlow, PyTorch, Scikit-learn
Key Technologies: Pandas, NumPy, OpenCV, AWS, Google BigQuery, Deep Learning, Performance Optimization, NLP

Experience

Research Intern — Indian Institute of Technology (IIT) Bombay, Mumbai, India Feb 2024 – Sept 2024

- Developed a **PHP and JavaScript–based** data visualization and theme builder using the **Yii Framework**, enabling 20+ professors to design and manage analytics dashboards and custom layouts.
- Built a full-stack **Figma-to-HTML parser** with drag-and-drop functionality and a **reusable data reporting theme system** for 100+ reports, supporting structured data presentation and visualization.
- Optimized **MySQL database** queries with **indexing strategies** and integrated **AJAX** for asynchronous data fetching, managing 500+ layout configurations; conducted **code reviews** with collaborating engineers to enforce quality standards.

UI/UX Design Intern — HTS Tech Solutions, Pune, India Jun 2022 – Dec 2022

- Designed **interactive data visualization** dashboards for analytics applications, applying **UI/UX design** principles.
- Created **wireframes** and **prototypes**, iterating based on feedback and ensuring **accessibility** standards compliance.

Projects

Satellite Image Segmentation Pipeline — SAM3, PyTorch, NumPy, Python Oct 2025 – Present

- Built a **semantic and instance segmentation pipeline** using **SAM3 (Segment Anything)** to automatically extract structural features (buildings, roads, water) from **high-resolution satellite imagery**.
- Engineered a **custom evaluation framework** in **PyTorch and NumPy** computing **per-class IoU and pixel accuracy**, achieving **mIoU of 0.62** and **pixel accuracy of 0.78** across 17 test images.
- Optimized pipeline for **GPU memory efficiency** using inference-mode execution and **CUDA cache management**, resolving **out-of-memory errors** on large-scale satellite images.

RAG-Powered Code Assistant — LangChain, ChromaDB, CMake, Docker, CI/CD March 2026

- Built a **Retrieval-Augmented Generation (RAG)** system using **LangChain** and a local **LLM (Ollama/Llama 3)**, enabling semantic search across 5,000+ documentation pages via a **ChromaDB vector database**.
- Packaged for **PyPI** and **Conda** using **scikit-build-core** with cross-platform Linux/Windows support; designed a **GitHub Actions CI/CD** pipeline for automated multi-OS builds, containerized with **Docker**, and achieved **85% test coverage** via **pytest**.

ASL Detection Model — TensorFlow, Docker, AWS, GitHub Actions July 2024

- Designed a **deep learning CNN** for **American Sign Language recognition** with OpenCV preprocessing; deployed containerized model on **AWS EC2** via **Docker** with a **GitHub Actions CI/CD** pipeline for automated testing.

Achievements

- **Publications:** First author, Springer (Automated GUI Testing, UX); co-author, IEEE (Blockchain for Disaster Relief).
- **Leadership:** Led UI/UX team at OIR, VIIT, coordinating projects involving **data visualization and user analytics**.
- **International Exposure:** Completed Summer & Spring School on **Advanced Engineering and Data Science**, Tokushima University, Japan.